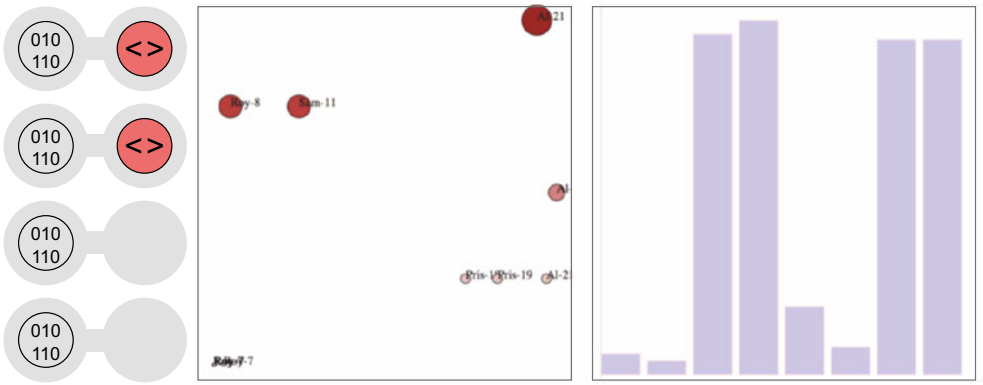


Figure 2.1 Examples from this chapter, including a diagram of how data-binding works (left) from section 2.3.3, a scatterplot with labels (center) from section 2.3, and the bar chart (right) we'll build in section 2.2

2.1 Working with data

We'll deal with two small datasets in this chapter and take them through a simplified five-step process (figure 2.2) that will touch on everything you need to do with and to data to turn it into a data visualization with D3. One dataset consists of a few cities and their geographic location and population. The other is a few fictional tweets with information about who made them and who reacted to them. This is the kind of data you're often presented with. You're tasked with finding out which tweets have more of an impact than others, or which cities are more susceptible to natural disasters than others. In this chapter you'll learn how to measure data in D3 in a number of ways, and how to use those methods to create charts.

Out in the real world, you'll deal with much larger datasets, with hundreds of cities and thousands of tweets, but you'll use the same principles outlined in this chapter. This chapter doesn't teach you how to create complex data visualizations, but it does explain in detail some of the most important core processes in D3 that you'll need to do so.

2.1.1 Loading data

As we touched on in chapter 1, our data will typically be formatted in various but standardized ways. Regardless of the source of the data, it will likely be formatted as single-document data files in XML, CSV, or JSON format. D3 provides several functions for

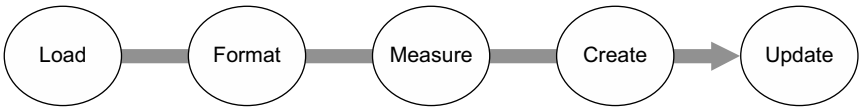


Figure 2.2 The data visualization process that we'll explore in this chapter assumes we begin with a set of data and want to create (and update) an interactive or dynamic data visualization.